



ID verification in Tinder

Tinder will make ID verification available to members to boost safety.
<https://digit.in/sep21-79>



Audi's new concept car

Audi reveals a concept car with shape-shifting capabilities.
<https://digit.in/sep21-80>

and you want to take it back in to get it repaired, that's one thing. But if you have an issue with your car and you have to take it into the shop it's a very expensive proposition. So it's not just about temperature extremes, it's also about quality and reliability. And the other thing that's quite a bit different that happens in the automotive space is the expectation around the platforms' longevity of supply and support. So several things relate to DPM (Defects Per Million) or quality, the overall reliability of the part, functional safety and how that fits into aligning with the host and the system solution, in regards to providing notification(s) of whether or not there's a potential to have a failure onboard. Or if you have a failure then how does the system react to that? So that's a lot different than you might see in a consumer application.

But aren't these sort of similar to what is expected at data centres where, let's say if you are working with precision calculations, then you need precision to let's say 16 bits, 32 bits or 64 bits, and you're looking at memory chips that can identify errors in its bits and provide that notification. So could you say it's more similar to building chips for data centre applications?

Kris: I think that some of those things are very similar, but beyond the defect expectations, there are different environmental conditions. The data centre environment is much more controlled. And in an automobile, you can be writing data at minus 40 degrees Celsius and reading it at 115 degrees Celsius. So you've got a much different temperature-cross requirement that's critical for the technology in automobiles.

On average in a well-equipped automobile, how much memory does it have, and how do you see that changing over the next 30 years?

Kris: It's completely different across the spectrum of vehicles. You know, you can have a vehicle that has very little memory in it. It still has

a mechanical dashboard. It has doesn't have a premium In-Vehicle Infotainment system or navigation system, and it doesn't have any sort of dash-related systems and so those can be pretty limited. Then you've got high-end systems that have a massive amount of storage in addition to a lot of DRAM memory. So you know you can go from having Gigabits to GigaBytes of DRAM and you can also go from having very little storage, say maybe 8 or 16 GigaBytes of memory storage to having a TeraByte of storage in the system. So there's a pretty broad range of vehicles and different expectations in that space.

And as automobiles get smarter, you have one aspect being that you'd need a lot more onboard memory for calculations and storage. Therefore, let's say if it's a Tier 5 autonomous



vehicle, then you're looking at petabytes and exabytes about 5 to 10 years down the line. But at the same time, we're also going to see innovations on the communication front, where data that is not critical and does not require to be immediately processed can be shipped off to an edge node, which is later processed at a remote data centre and sent back to you.

Do you see the capacity of memory on vehicles increasing consistently, or do you expect it to sort of increase to a certain limit and plateau off because the communication technologies are better at that point?

Kris: Let me share with you just a little bit about how we see the memory in automobiles today. If you think

about it, there's probably four or five primary systems that require memory in storage, first and foremost, and the largest driver of memory today in storage today is the infotainment navigation system that's inside the vehicle. Next would be the digital dash. Then there are ADAS (Advanced Driver-Assistance Systems) systems themselves which will lead to autonomous vehicles in the future, and then there are powertrain systems. So that includes engine control systems and then the last one is this telecommunications that you're referring to. Yes, there is for sure more and more communications happening, but at the same time, we also see the expectation for onboard content storage. Whether or not it's applications or videos or movies or those sort of things. And this is all from a storage perspective.

At the same time, we we're seeing increasing requirements for things like - as you get more and more autonomous and include ADAS features - data recording, so you can think of it almost like a black box that you have in an aircraft that drives more and more local storage content to the vehicle versus things stored in the cloud. The memory

concept for the telecommunication system and applications themselves is pretty small. You can think of a radio that's not too different than what we had in some of the phones in the past, so they're not driving a ton of memory requirements. But as you pointed out, with 5G the ability to get access to information, what you store locally versus what you store in the cloud and download from the cloud will change that model. But we still see 3X growth in memory and storage happening over the next four to six years or so. We see significant growth and we don't see a significant roll-off associated with memory and storage being outside the vehicle. Those things will happen but we still see significant growth opportunities. **▶**